LINUX COMMAND CHEAT SHEET

Purpose: A practical reference for Linux navigation, file handling, permissions, process checks, network troubleshooting, logs, scheduling, and basic software management. Use these commands only on systems you own or are authorized to administer.

1. Moving Around and Managing Files

Linux Command	What It Does	When to Use It
pwd	Prints the path of the current directory.	Confirm the directory you are working in.
ls -lh	Lists directory contents with human-readable file sizes.	Review files and their sizes.
ls -A	Shows hidden entries except . and ..	Find configuration files without listing the two navigation entries.
cd ..	Moves to the parent directory.	Move up one level in the directory tree.
mkdir -p projects/week1	Creates nested directories as needed.	Prepare a folder structure in one command.
touch notes.md	Creates a file if it does not exist.	Create a placeholder for notes.
cp -i source.txt copy.txt	Copies a file and prompts before overwriting.	Make a copy while reducing accidental overwrites.
mv report.txt archive/	Moves a file to another directory.	Organize completed reports.
rm -i old.txt	Prompts before removing a file.	Delete a file more cautiously.
file document.bin	Identifies the likely file type.	Inspect an unfamiliar file.

2. Reading and Searching Text

Linux Command	What It Does	When to Use It
cat notes.md	Prints a file to the terminal.	Read a short text file.
less /var/log/syslog	Opens a scrollable viewer for a text file.	Review a large log file.
head -n 15 file.txt	Prints the first 15 lines.	Preview a file's beginning.
tail -n 25 file.log	Prints the last 25 lines.	Review recent log entries.
grep -n 'warning' app.log	Finds matching lines and includes line numbers.	Locate warning messages in a log.
grep -Rni 'failed' ./logs	Searches files recursively, ignoring case, with line numbers.	Search a practice log directory.
find . -type f -name '*.conf'	Finds regular files matching a name pattern.	Locate configuration files under the current directory.
sort names.txt	Sorts lines in a text file.	Put names or output into sorted order.
uniq -c sorted.txt	Counts adjacent duplicate lines.	Count repeated lines after sorting input.
wc -w report.txt	Counts words in a file.	Estimate the word count of a report.

3. Accounts and File Permissions

Linux Command	What It Does	When to Use It
whoami	Prints the current username.	Check the active account.
id username	Shows a user's UID, primary group, and supplementary groups.	Inspect an account's group membership.
getent group developers	Looks up a group entry.	Verify that a group exists and inspect membership.
sudo useradd -m learner1	Creates a user and a home directory on systems using useradd.	Set up a lab account with administrative authorization.
sudo groupadd labteam	Creates a group.	Create a group for shared lab access.

Linux Command	What It Does	When to Use It
sudo usermod -aG labteam learner1	Adds a user to a supplementary group.	Grant group membership; the user may need to sign in again.
chmod 640 report.txt	Owner can read/write; group can read; others have no access.	Limit access to a document.
chmod g+rw shared.txt	Adds group read/write permission.	Allow a group to collaborate on a file.
sudo chown learner1:labteam shared.txt	Changes file owner and group.	Set ownership for shared resources.
umask	Displays the current default permission mask.	Understand default permissions for newly created files.

4. Processes and Resource Checks

Linux Command	What It Does	When to Use It
ps -ef	Lists processes in a full-format view.	Inspect processes running on the system.
pgrep -a ssh	Finds processes matching a name and shows their command lines.	Check for an SSH-related process.
top	Shows live process and resource statistics.	Investigate CPU or memory pressure.
free -h	Reports RAM and swap usage in readable units.	Check memory availability.
df -Th	Shows filesystem space and filesystem types.	Check capacity and filesystem type.
du -h --max-depth=1 .	Summarizes the sizes of immediate subdirectories.	Find which folders use the most space.
uptime	Reports uptime and load averages.	Check system runtime and load.
kill -TERM PID	Requests a process to terminate gracefully.	Stop a process you are authorized to manage.

5. Network Inspection and Troubleshooting

Linux Command	What It Does	When to Use It
ip -br address	Shows a compact list of interfaces and addresses.	Record VM IP addresses clearly.
ip route show	Displays routes known to the system.	Find the default gateway.
ping -c 4 192.168.56.20	Sends four ICMP echo requests.	Test reachability to another lab VM.
tracepath example.com	Traces a path toward a destination where supported.	Investigate path or MTU issues.
ss -lnt	Lists listening TCP sockets.	Check which TCP ports are listening.
ss -lun	Lists listening UDP sockets.	Review UDP listeners.
getent hosts example.com	Uses the system's configured name-resolution sources.	Check how the host resolves a name.
dig +short example.com	Prints a concise DNS answer.	Quickly inspect DNS resolution.
curl -sS -I https://example.com	Requests response headers and reports connection errors.	Check HTTP response metadata.
ssh learner@192.168.56.20	Connects to a remote host using SSH.	Log in to an authorized lab machine.

6. Logs, Services, and Scheduling

Linux Command	What It Does	When to Use It
ls -lah /var/log	Lists log files with details.	Find logs available on the system.
journalctl -p warning -b	Shows warning-level and higher journal messages from the current boot.	Look for system warnings.
systemctl is-active cron	Checks whether the cron service is active on systems using that unit name.	Confirm the scheduler service status.
systemctl --type=service --state=running	Lists running service units.	Review active services.
crontab -e	Opens the current user's cron schedule for editing.	Create a recurring task.
crontab -l	Lists the current user's scheduled jobs.	Verify a cron entry.

Linux Command	What It Does	When to Use It
date '+%F %T %Z'	Prints date, time, and timezone in a consistent format.	Record timestamps for lab notes.

7. Archives, Documentation, and Software

Linux Command	What It Does	When to Use It
tar -czf logs-backup.tar.gz logs/	Creates a gzip-compressed tar archive.	Package a directory for backup.
tar -tzf logs-backup.tar.gz	Lists files inside a gzip-compressed tar archive.	Inspect an archive before extracting it.
tar -xzf logs-backup.tar.gz	Extracts a gzip-compressed tar archive.	Restore archived files.
man grep	Opens the manual for grep.	Learn supported options and usage.
grep --help	Prints command help.	Quickly check available flags.
command -v python3	Shows how the shell resolves a command name.	Check whether a program is available on PATH.
sudo apt update	Refreshes APT package indexes on Debian/Ubuntu.	Prepare to install software.
sudo apt install traceroute	Installs traceroute on Debian/Ubuntu systems where the package is available.	Add a routing diagnostic utility for lab exercises.

8. Permission Numbers at a Glance

Linux permissions are assigned separately to the owner, group, and others. For a regular file, read permits viewing, write permits modifying, and execute permits running. For a directory, execute permits traversal.

Number	Symbol	Meaning
7	rwx	Read, write, execute
6	rw-	Read and write
5	r-x	Read and execute
4	r--	Read only
0	---	No access

Example: chmod 750 team-folder gives the owner full access, the group read/traverse access, and others no access.

Example: sudo chown learner1:labteam team-folder changes the folder's owner and group. Be careful with recursive changes and never use chmod -R 777 as a shortcut.